

112B Data Science and Python Programming

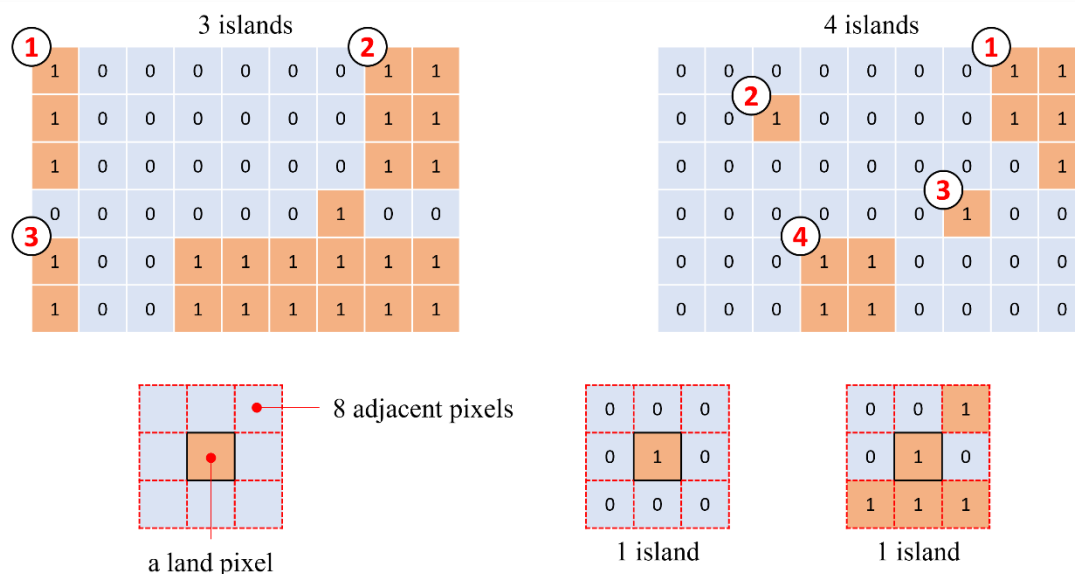
Homework Assignment #1

Due: 2024/3/22 12:00:00

According to the tenet of **Pastafarianism**, the central creation myth is that an invisible and undetectable Flying Spaghetti Monster created the universe after drinking heavily. According to these beliefs, the Monster's intoxication was the cause for a flawed Earth. Fortunately, you are sober now (maybe not?). In this final project, we are going to create a simple world that only contains land and water. Your task is to count how many islands do we have in this simple world. Enjoy it!

Problem #1: Number of islands (60%)

Please use the function “**create2Dmap**” in “**utils.py**” to create a 2D array with random elements 0 or 1. Assuming 0 is water and 1 is land in this simple world. Any adjacent land forms an island. Please write a program to count how many islands do we have in this world. Here is the rule:



Here is the sample code:

```
class solution_hw1:
    def __init__(self, img):
        self.original_img = img
        self.img_copy = np.copy(img)

    def numIslands(self):
        """Method for counting"""
```

```

    ???

def search(self, input, i, j):
    """Method for searching"""
    ???

```

If everything goes well, you can test your solution on the data “test_data.npy.” Your code might get this result:

```

(nycudopcs) 0:\                                     \112B\HW1>python main_hw1.py
# of Islands in Data 0: 13
# of Islands in Data 1: 9
# of Islands in Data 2: 8
# of Islands in Data 3: 7
# of Islands in Data 4: 13
# of Islands in Data 5: 11
# of Islands in Data 6: 18
# of Islands in Data 7: 13
# of Islands in Data 8: 15
# of Islands in Data 9: 7

```

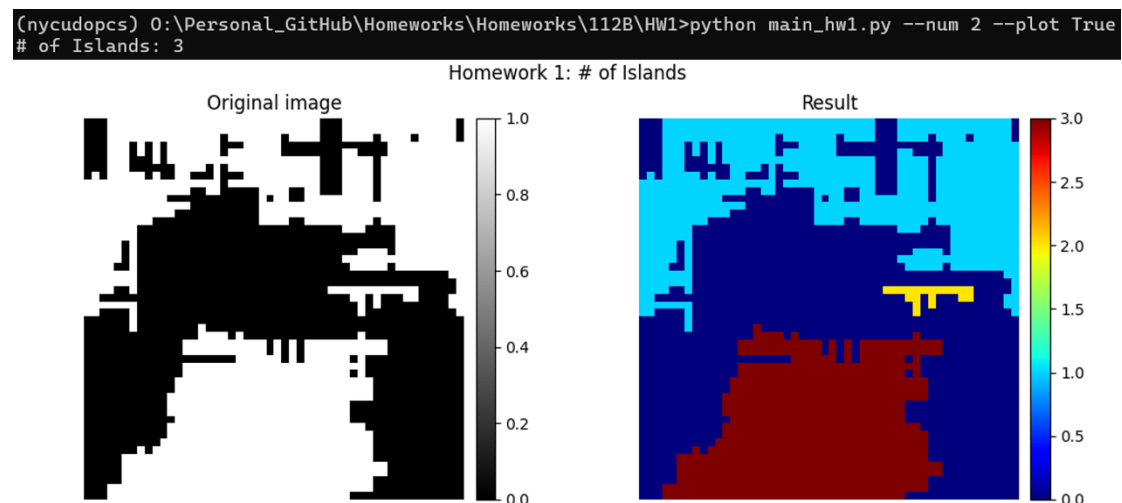
FRIENDLY REMINDER

Be careful with the directory of “test_data.npy”

Problem #2: Plot the result (40%)

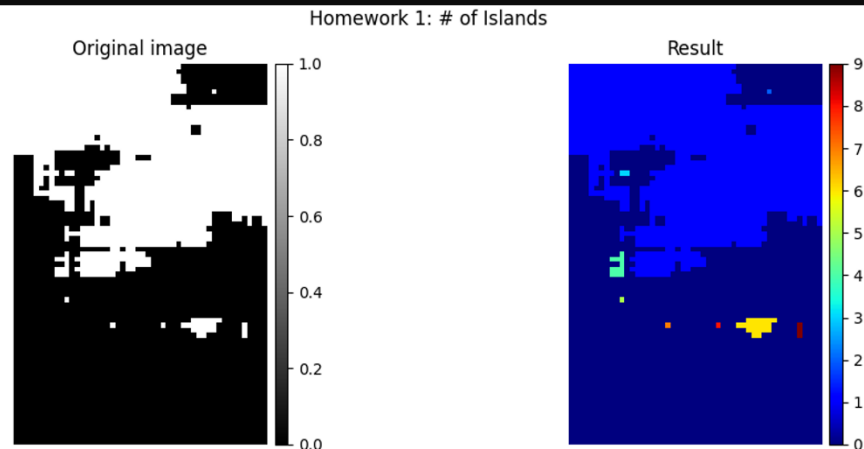
Please create a 2D array with arbitrary size, then plot your result. If everything goes well, your code might get these results:

1. Image size = (50,50), kernel size = (15,15)



2. Image size = (75,50), kernel size = (21,31)

```
(nycudopcs) 0:\
--size 75 50 --k 21 31
# of Islands: 9
.112B\HW1>python main_hw1.py --num 2 --plot True
```

**FRIENDLY REMINDER**

You don't need to plot a figure that is exactly same as mine.

Please accomplish this homework with an organized code (e.g., with main script and other script). Here is a template for your code structure:

```
112B_hw1_0123456789
├─ func.py           # Solution of hw1
├─ utils.py          # Utility functions for hw1
├─ ooo.py             # Package ooo for hw1
├─ xxx.py             # Package xxx for hw1
├─ ...                # More if necessary
└─ main_hw1.py        # Main scripts of hw1
```

The filename of your main script should be “**main_hw1.py**.”

You don't need to follow this structure, just keep your main script clean.

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs (e.g., signature). **Your signature should be attached in your main script.** Here is the template of signature:

```
"""  
Created on Sun Aug 7 01:23:45 2022  
  
@author: Xi Winnie, student ID  
  
@collaborators: Jane Doe, her student ID  
                John Doe, his student ID  
"""
```

Please save your code as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

112B_hw1_**ID**.zip

For example,

112B_hw1_**0123456789**.zip

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “E3@NYCU” system. There will be a section for uploading your homework.